

Robert Catalin Crisan

Software Engineer · Gameplay Programmer · Web Developer

robert.crisan2106@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PERSONAL PROFILE

A passionate and technically versatile developer with a strong foundation in C# and C++, built through hands-on experience across games development, software engineering, and interactive web projects. Skilled at architecting complex systems from the ground up, whether that means designing AI agent behaviour, engineering networked multiplayer experiences, writing GPU shaders, or building clean, maintainable application logic. Comfortable working independently or as part of an Agile team, with a track record of delivering under pressure and a drive to keep growing technically in any environment.

KEY SKILLS

Languages: C# · C++ · HLSL / GLSL · JavaScript · HTML / CSS

Engines & Tech: Unity Engine · Unreal Engine 5 · XR Interaction Toolkit (XRIT) · Normcore VR

Engineering: Gameplay Programming · OOP · Network Replication · GPU Shader Math · Agent-Based AI (FSMs, Pathfinding) · Physics & Maths for Game Engines · Data Modelling & Simulation Design

Methodologies: Git / GitHub · Agile / Scrum · Technical Documentation · Asset Management

Soft Skills: Analytical Problem-Solving · Systems Thinking · Cross-Disciplinary Collaboration · Rapid Prototyping

TECHNICAL PROJECTS

Adaptive AI Crowd Simulation

Unity · C#

Final Year Individual Project

- Designed an AI-powered crowd simulation using Finite State Machines (FSM) to model decentralised awareness and peer-to-peer data transmission under emergency stress scenarios.
- Engineered a custom NavMesh pathfinding system with dynamic hazard recalculation and raycast wall-avoidance vectors.
- Implemented an Explainable AI (XAI) data-logging architecture to capture real-time decision triggers and evaluate simulation performance.
- Optimised system performance using GPU-accelerated VFX graphs and queue management throttling to maintain high framerates under heavy load.

Echoes of the Twin Tombs

Unity · C# · Normcore

XR Gameplay Engineer & Art Director

- Developed an asymmetrical, networked cooperative VR escape room using the XR Interaction Toolkit for physical, hands-on player inputs.
- Engineered physics-based spatial mechanics including a recursive raycasting system to track continuous laser beams across dynamically rotating mirrors.
- Implemented seamless network replication and state synchronisation via Normcore, achieving zero-latency manipulation and smooth rotation interpolation across client instances.
- Built randomised logic algorithms with robust fail-safes to guarantee unique puzzle states per session without breaking game flow.

E.R.O. (Experimental Reconstruction Organism)

Unreal Engine 5 · C++

Co-op Open World Survival - Agile Team Project

- Developed a large-scale multiplayer open-world project within an Agile/Scrum team, implementing core gameplay loop systems from design through to stable release.
- Led server-client network replication, ensuring robust synchronisation across player actions, session stability, and a full inventory lifecycle.
- Conducted rigorous peer code reviews and logic revisions to maintain feature stability and overall code quality.

Custom Shader Programming

Unreal Engine 5 · HLSL

Technical Art Project

- Programmed technically advanced procedural shaders for an immersive environment, applying complex mathematical and algorithmic thinking to graphical data transformation.
- Engineered highly reactive environmental elements including procedurally generated fire and a dynamically dilating graphical eye driven by player proximity.

Peter Defeater - 2D Shooter

Unity · C#

Object Oriented Programming

- Programmed a progressively difficult 2D shooter demonstrating strict OOP principles: polymorphism, encapsulation, and inheritance.
- Developed dynamic character controllers, raycast-based collision detection, and custom MonoBehaviour / non-MonoBehaviour class hierarchies to manage game flow and weapon upgrades.

Rapid Prototyping & Game Jams

Unity · C#

Moderation · Small Boat · Mask

- Built three functional game prototypes under strict 48-hour constraints, rapidly implementing core scripts, weapon systems, and custom UI logic.
- Maintained project stability through strict Git version control, cross-discipline asset management, and collaborative pair programming.

EDUCATION

BSc (Hons) Games Development

2023 – 2026

University of Greenwich, London · Expected Grade: 1st Class Honours

- Core modules: AI & Intelligent Systems, Mathematics for Computing, Object-Oriented Programming, Software Engineering, Simulation Design.
- Final year project demonstrates the ability to build technically sophisticated, real-world applicable software systems.

ADDITIONAL INFORMATION

Right to Work: Full right to work in the UK. | **Driving Licence:** Full UK Category B. | **Availability:** Available for full-time roles. Willing to commute or relocate within the UK.